

₁ Mechanistic Explainability in Reinforcement
₂ Learning

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Chapter 1

Conclusion

This manuscript set out to explain the behavior of Reinforcement Learning (RL) agents by analyzing the internal representations of their Neural Networks (NNs). This final chapter takes stock of this work. It begins by revisiting each research question in light of the contributions presented. It then discusses the limitations of these contributions. Finally, it presents the directions for future research that these limitations open.

1.1 Summary of Contributions

This section revisits the four research questions introduced in Section ?? . For each of them, it summarizes the answer provided by this manuscript.

Research Question ?? concerned the approaches proposed in the literature to analyze the internal workings of NNs. To answer it, we first made explicit the working hypothesis shared by most of this literature, the *Concept Hypothesis* (Hyp. ??). According to this hypothesis, NNs generalize by organizing their Latent Space (LS) around concepts. We then organized existing methods into four families: probing, interpretable directions, Sparse Autoencoders (SAEs), and clustering. Each family studies the LS from a different angle, but they share the goal of understanding how Latent Representations (LRs) are structured.

Research Question ?? concerned the organization of existing eXplainable Reinforcement Learning (XRL) methods [Alaarabiou et al., 2024, Alaarabiou et al., 2026b]. To answer it, we first clarified what an explanation is by placing the observer at the center of the explanatory process. Building on this frame, we identified the subjects and sources of explanation specific to RL. The subject of explanation is the policy of the agent, which can be explained either at the level of a single decision or at the level of its strategy over time. The sources of explanation are the policies, the trajectories, and the value functions. From these elements, we proposed a taxonomy that accounts for the specificities of RL and used it to review the state of the art.

Research Question ?? concerned the use of LRs to identify the mechanisms underlying the decisions of RL agents. To answer it, we introduced Clustering Agent Representation (CAR), an XRL method based on the principles of Mechanistic Explainability (ME) [Alaarabiou et al., 2026a]. CAR trains several surrogate policies independently to reproduce the same agent. It then clusters

their LRs and compares the resulting partitions. This procedure relies on the *Representation Convergence Hypothesis* (Hyp. ??), according to which independently trained networks develop similar concepts for a given task. Applied to six Atari environments, CAR recovers similar clustering structures across surrogate policies. The qualitative analysis shows that these clusters capture interpretable aspects of the task, such as episode progression, specific game situations, and behavioral patterns.

Research Question ?? concerned the objective quantification of explanation quality. To answer it, we introduced three metrics. The *Clustering Universality* (CU) measures the consistency of the clusters across independently trained surrogate policies. The *Abstraction Score* (AS) measures how far these clusters depart from the structure of the observations, of the actions, and of an untrained network. The experiments show that these two properties evolve in opposite directions across the network. The earliest layers are the most consistent, but also the least abstract. The *Adjusted Clustering Universality* (ACU) combines both properties into a single measure and provides a practical criterion for selecting the layer to analyze. These metrics make it possible to compare competing hypotheses about the structure of a LS without relying on manual inspection.

Taken together, these contributions form a coherent chain. The *Mechanistic Hypothesis* (Hyp. ??) states that NNs can be explained through their internal mechanisms. The *Concept Hypothesis* (Hyp. ??) states that these mechanisms are organized around concepts. The *Representation Convergence Hypothesis* (Hyp. ??) states that these concepts reflect the task. CAR translates these hypotheses into an empirical procedure, and the experiments provide support for them.

1.2 Limitations

Although the research presented in this thesis has produced encouraging results, it remains subject to limitations. This section discusses the four main limitations identified at this stage.

First, CAR describes how the LS of an agent is organized, but not what each cluster means. The metrics quantify the consistency and abstraction of the clusters, but they say nothing about their content. In this manuscript, the meaning of each cluster was assigned through manual inspection of the corresponding observations. This step is time-consuming, subjective, and difficult to scale to a large number of clusters or environments.

Second, the clusters identified by CAR are not yet directly related to the decisions of the agent. We observe that the agent distinguishes certain situations, but not how these distinctions lead to a given action. As a result, CAR cannot yet explain a specific failure or reveal a bias in the behavior of the agent.

Third, the experiments were conducted on six Atari environments. These environments are diverse, but they remain limited in size and complexity. Whether CAR scales to more complex environments and larger networks has not yet been established. In addition, only one clustering method was evaluated, based on k -means with Euclidean distance. Other distances and other clustering methods could reveal different structures in the LS. Since the pipeline of CAR is modular, these alternatives can be tested and compared using the proposed metrics.

Finally, the proposed metrics have only been applied to clustering. The

other families of methods reviewed in Chapter ?? have not yet been evaluated within this framework. This is particularly the case for methods based on the *Superposition Hypothesis* (Hyp. ??). These methods are widely used, but their assumptions remain debated. Our metrics could be used to test these assumptions quantitatively, and to compare these methods with CAR on the same tasks.

1.3 Perspectives

The limitations discussed above open several directions for future research. This section presents three of them.

A first direction is to automate the semantic interpretation of the clusters. One approach is to rely on vision-language models. Given observations from the same cluster, such models could generate a natural language description of what distinguishes it from the others. This approach would provide a textual description for each cluster, helping identify its specific semantics and fully automate the CAR method.

A second direction is to use the identified clusters to explain the strategy of the agent. This manuscript has shown that the LS can be used to identify meaningful concepts. The next step is to use these concepts to describe how the agent behaves over time. Since each cluster groups states that are similar from the agent's perspective. The transitions observed between these clusters during episodes would then form a simplified model of the environment, as perceived by the agent.

A third direction concerns the evaluation of explanations. The metrics proposed in this manuscript measure structural properties of the LRs. They are necessary, but they are not sufficient. According to our definition, an explanation is valuable only if it improves the observer's understanding of the agent. Measuring this improvement requires experimental protocols from disciplines such as cognitive science and psychology. Such protocols could assess whether an explanation actually improves the observer's mental model of the agent. This is the ultimate goal of explainability, and evaluating it would complete the answer to Research Question ??.

117 Acronyms

118	ACU <i>Adjusted Clustering Universality</i> 2
119	AS <i>Abstraction Score</i> 2
120	CAR Clustering Agent Representation 1–3
121	CU <i>Clustering Universality</i> 2
122	LR Latent Representation 1–3
123	LS Latent Space 1–3
124	MDP Markov Decision Processes 3
125	ME Mechanistic Explainability 1
126	NN Neural Network 1, 2
127	RL Reinforcement Learning 1
128	SAE Sparse Autoencoder 1, 3
129	XRL eXplainable Reinforcement Learning 1

Bibliography

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